

PHASE 01

Heat up + Calibration

QUICK OPERATOR GUIDE
Release 0.9.36 · Build 2026.08.11-r15

PURPOSE

Heat the CK-1 evaporator, monitor both QMBs and calculate the CK-1/Sample thickness ratio required by Phase 03.

WORKFLOW AT A GLANCE

1 · HEAT Keysight automatic ramp	2 · READY CK-1 T and average rate at target	3 · OPEN Open shutter physically and confirm	4 · CALIBRATE Track relative QMB thickness	5 · FINISH Close shutter, ramp to 0 A, output OFF
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BEFORE STARTING

- ✓ Confirm chamber pressure, shutter position and QMB signals.
- ✓ Switch on the Keysight and verify the expected protected state.
- ✓ Check the temperature and rate targets before starting the run.

KEY FACTS

Shutter gate	CK-1 temperature $\geq$ target and average CK-1 rate $\geq$ target.
Automatic current cap	0.660 A default
Software hard stop	0.680 A
Temperature watchdog	255 C default

CONTROLS AND END STATES

Finish Phase	Normal completion: controlled ramp-down to 0 A, then OUTPUT OFF.
Abort / Safe Stop	Emergency stop: immediate CURR 0.000 and OUTPUT OFF; closes Phase 01 only.

RUN DATA

Saved under:
Data Samples/Heat up + Calibration Data/

SAFETY - VERIFY THE REAL HARDWARE

Abort removes evaporator power before saving files. Controller slope, trend and cascade diagnostics do not add hidden shutter conditions.